Contents

The Completion

A relativistic field theory carrying $a_0 = \kappa c \sqrt{}(G\rho\_\Lambda)$

Zimmerman, Carl P. — Briar Creek Tech Assembled 2026-08-09; **v2, v3, v4, v5 same day; v6 2026-08-10; v7 2026-08-10 (late)**. **v7 changes — the largest POSITIVE structural change of the programme: a_0(z) is DERIVED from the action.** Until v7 the redshift scaling of a_0 was imposed on the theory as a phenomenological law dressed in DESI's CPL parameters — a dressing that was **never self-consistent** with this paper's own w = -1 exact (§1.2), and is here withdrawn and replaced. The derivation (new §1.3, rows 18–19): promote the constant a_0^2 to **$a_0^2(Q) = \kappa^2 c^2 G(-K(Q))/c^2$ — the MOND scale is the dark sector's PRESSURE** (p = K is an identity in this sector). Zero new parameters; ONE new relation $\mu^2 \Lambda D^2 = M^4$ ***("the Lagrangian vanishes at the DBI wall"), which* removes *a free parameter (§3: 5 → 4). Today -K = ρΛ:* the committed coefficient to the digit. Into the past the trace excitation's positive pressure climbs the DBI wall and cancels the vacuum's negative pressure: MOND switches off at recombination** as an output**, with w = -1 still exact — the vacuum never rolls; the *total pressure* evolves. An adversarial external pass (Gemini) proved the density promotion ($a_0^2 \propto \rho Q$) *yields a_0 = const or the wrong sign; that theorem is what forces the pressure route, and is credited in row 18. The full perturbation matrix with the new* YQ mixing is DONE (row 19): the cosmological matrix is untouched *exactly*, the no-ghost theorem **transfers** kernel-independently, c_T = 1 survives. Also in v7: the forest-sector correction — stage 15's claim that the Hooper et al. generic-shape likelihood is public was WRONG (announced, never released; all probes 404); the sector's forest suppression is instead **self-computed** by a lognormal mock (0.20% at the binding bin, nuisance-dominated, `stage16`). Still owed after v7: the CLASS re-run with the derived law, the MUSE/MSA-3D re-exam against the bumpless shape, and a *derivation* of $\beta = 1$ (it is **selected**, not derived — non-claim 2e). **v6 changes — an ERRATUM against v5, filed against my own argument.** v5's non-claim 2d rested one of its three steps on a **category error**: it bounded the *export* of the dust's energy from a galaxy by the khronon's own polytropic sound speed (≈ 690 Gyr). *∇μT^μν = 0 bounds the **total energy**, not the **flux velocity**; the bound on how fast energy can leave a region is **causality**, and AeST has two massless tensor modes at exactly* c *— by this paper's own row 17. An O(*c*) export channel therefore exists in the theory as written (1 Mpc in 3.3 Myr).* ***That step is withdrawn.*** *Also withdrawn: the claim that the leak rates "lock" at* ρΛ/ρ_dust = 2.60 — that is merely today's density ratio. **The conclusion survives on stronger, published, transport-independent grounds:** exporting the energy removes it from the *galaxy*, not the *universe*, so the measured global budget decides — a halo-gated conversion needs a converted fraction $\zeta \approx 0.6$ against the **epoch-marginalised** bound $\zeta < 0.0204$ (Planck) to 0.0374 (+lensing+BAO+SN+DES) [McCarthy & Hill, PRD **108**, 063501], over by 16–29×, and would leave Ω*dr,0 ≈ 0.196 ≈ 2100*×

today's radiation density; the dust→dark-energy channel needs far more than the published $|\delta_0| \leq 6.7 \times 10^{-4}$. And the halo gate is ***adverse****, not protective: dilution goes as a_c, so later injection leaves* more *residue.* ***The lensing theorem (ρ+3p vs ρ+p) is untouched and is what carries the result.*** *Banked as a genuine asset: the Y-gate's FRW exactness makes recombination-era N_eff immunity* exact. *Also recorded, against expectation: the* ***background*** *does not require dark matter at all (Kunz's dark degeneracy, confirmed numerically — a single free w_X(z) reproduces ΛCDM's H(z) with baryons alone); the binding constraint is* ***CMB lensing****, which forces the clustering to survive to z < 0.30. Every numbered property below is backed by a committed, runnable script that exits non-zero on failure. Scripts named at each line.* ***v5 changes — the largest revision, and it runs against the programme's interest.*** *A six-stage sequence (`nbody_2026/`, every stage committed and green) settles the galaxy-interior question that v3 reopened as non-claim 2d, and* ***settles it adversely: the Q-sector cannot also be absent from galaxies, and the repair space of local, energy-preserving modifications is provably empty.*** *Three results carry it: (i)* the dust mass IS the conserved shift charge *($\rho = Q_0 n$ with $\rho/n = Q_0$ regardless of any added Y-dependent Q-mass), so it cannot be suppressed locally — which retro-explains three withdrawn mechanisms; (ii)* unlocking the charge does not unlock the energy *(*$\nabla\mu$T^$\mu\nu$ = 0 holds regardless — though v6 withdraws the transport step that accompanied this, see the v6 banner); (iii) ** dynamics responds to ρ + 3p but lensing responds to ρ + p, and no equation of state kills both** — so the same no-slip identity that underwrites row 1 forbids hiding the dust. Also corrected here: v3/v4's claim that the DBI cap bounds the dust *density* (it bounds the pressure; *ρexc is linear in u and diverges at saturation), and v4's spliced amplitude band.* ***Rows 1–12 and 15–17 are untouched. v2 changes:*** *row 13's shift-charge cluster route is* ***withdrawn*** *(killed by the 1-Mpc confrontation: smooth accretion restores* $\$ \to 1 foranycoldIC)andreplacedbythe * *a\{0\}-bumpenvironmentresponse * *, whichpassedafive-environmentmatrixANDitsperturbationhealthcheck; thehealthcheckcarvedtwohardconst$
$\leq 8.4 \times 10^{-7}$; amplitude $\leq 2.7\times$ fiducial). The coefficient row now reads as **measured**: $\kappa = 0.551 \pm 0.043$, $\pm$ 0.063 if the H_0 tension is carried as a systematic ($\kappa \propto a_0/H_0$). **v3 changes:** (i) the **full AeST perturbation matrix** for the bump term is now DONE and added as row 17 — c_T = 1 exact, no-ghost as a theorem, quasi-static closure derived, amplitude cap softened to a band; non-claim 2c is updated accordingly. (ii) **Non-claim 2 is corrected**: v2's "by item 13 it is absent where rotation curves are measured" was a dangling pointer — row 13 was withdrawn *in the same version* — so the galaxy-interior status of the Q-sector dust is **REOPENED** and now stated as its own open problem (non-claim 2d). v2 carried this internal inconsistency for several hours; the correction runs against the framework's interest and is flagged rather than hidden. (iii) §3's amplitude cap now states its units (4.5 Mpc^{-2} = 2.7× the fiducial 1.65 Mpc^{-2}).

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1. The action

$$S = \int d^4x\,\sqrt{-g}\,\Big[\ \underbrace{\frac{R - 2\Lambda_{\text{bare}}}{16\pi G}}_{\text{Einstein–Hilbert}} \ +\ \underbrace{\mathcal{L}_{\text{aether}}[A^\mu, g]}_{\text{unit-timelike } A^\mu A_\mu = -1} \ +\ \underbrace{\mathcal{F}(Y, Q)}_{\text{the dark sector}}\ \Big] \ +\ S_{\text{matter}}[g, \psi]$$

with the two scalar invariants built from the khronon φ and the aether A^μ,

$$Q \equiv A^\mu \nabla_\mu \varphi, \qquad Y \equiv (g^{\mu\nu} + A^\mu A^\nu)\nabla_\mu\varphi\,\nabla_\nu\varphi.$$

This is **Aether-Scalar-Tensor** (Skordis & Złośnik 2021, PRL **127** 161302) — chosen because it is the only relativistic MOND-class theory that reproduces the CMB. The framework's content is entirely in the choice of the free function , which **factorises**:

$$\boxed{\mathcal{F}(Y,Q) \;=\; \underbrace{\frac{a_0^2}{8\pi G}\,\mathcal{F}_Y\!\left(\frac{Y}{a_0^2}\right)}_{\text{MOND}} \;+\; \underbrace{K(Q)}_{\text{dark energy + dark matter}} \;+\; \underbrace{A\,B\!\left(\tfrac{Y}{a_0^2}\right)(Q-Q_0)^2}_{a_0\text{-bump response (v2)}}}$$

$$B(y) = \frac{y}{(1+y)^2}$$

The third term (v2) is a **position-dependent Helmholtz mass** $\mu_{\text{eff}}^2 = A\,B(g^2/a_0^2)$ **peaked at the framework's own** a_0 — the location costs nothing (Y/a_0^2 is already the Y-sector's normalisation); one new calibrated amplitude $A \approx 1.7\ \text{Mpc}^{-2}$. *The dark sector's response is resonant at the MOND transition, and clusters are the cosmic objects that live at* a_0 (R_{500} at 0.33–0.58 a_0).

1.1 The Y-sector — MOND

_Y is fixed by the framework's Route A kernel ν(y) = 1/(1 - e^(-$\sqrt{}$y)), y = g_bar/a$_0$. In AQUAL variables it is the closed parametric pair

$$\mu(u) = 1 - e^{-u}, \qquad x(u) = \frac{u^2}{\mu(u)}, \qquad u = \sqrt{y},\ x = g_{\text{obs}}/a_0.$$

with the normalisation

$$a_0 \;=\; \kappa\, c\sqrt{G\rho_\Lambda} \;=\; \frac{cH_\Lambda}{Z} \;=\; c^2\sqrt{\frac{\Lambda}{32\pi}} \;=\; 9.3619\times 10^{-11}\ \text{m}\,\text{s}^{-2}, \qquad \kappa = \tfrac{1}{2},\ \ Z = 2\sqrt{8\pi/3}.$$

$\kappa = \frac{1}{2}$ **is FITTED, not derived.** See §4.

1.2 The Q-sector — an offset DBI khronon

$$\boxed{K(Q) \;=\; -M^4 \;+\; \mu^2\Lambda_D^2\left[1 - \sqrt{1 - \frac{u^2}{\Lambda_D^2}}\,\right], \qquad u \equiv Q - Q_0, \quad |u| < \Lambda_D}$$

Three jobs from one function:

piece	limit	what it is
the **offset** -M^4	u = 0, K = 0	ρ = -K = M^4, p = K = -M^4 $\Longrightarrow$ **w = -1 exactly**: dark energy

piece	limit	what it is
the **deviations**	u ≪ Λ_D	K → μ^2u^2/2 is the **pressure**, while ρ_exc = Q$_0\mu^2$u is **linear** in u ⟹ w = u/2Q$_0$ → 0: **dust**, i.e. the dark matter. *(v5: the linearity is why the dust mass equals the conserved charge — see non-claim 2d.)*
the **saturation**	u → Λ_D	K bounded, K → ∞ ⟹ w → 0 again: kills the early stiff phase

Parameters: **M** (vacuum scale, M^4 = ρ_Λ), μ (Helmholtz mass), **Λ_D** (DBI field-space scale), **I**$_0$ (the conserved shift charge = the dust amount). **(v7: Λ_D is no longer free — μ^2Λ_D^2 = M^4, §1.3.)**

1.3 a$_0$(z) — derived, not imposed (v7)

The constant a$_0^2$ in §1.1 is promoted to a scalar of the Q-sector:

$$\boxed{a_0^2(Q) \;=\; \frac{\kappa^2 c^2\, G\,(\,-K(Q))}{c^2}\,, \qquad \text{i.e.} \qquad a_0 = \kappa c\sqrt{\frac{G\,(-p_Q)}{c^2}}}$$

The MOND scale is the dark sector's pressure — and since p = K identically (§1.2), this is the unique algebraic scalar of Q that (i) equals ρ_Λ today (u → 0: -K = M^4, reproducing §1.1's normalisation to the digit) and (ii) *declines* into the past. The competing candidate, a$_0^2$ ∝ ρ_Q, is **excluded by one bit**: the density grows with excitation, so a$_0$ would rise toward recombination — MOND ON at the CMB (the adversarial theorem credited in row 18 proved that route also gives a$_0$ = const on the pure-vacuum branch; either way it is dead). The pressure route additionally has charge-suppressed backreaction: da$_0^2$/dQ ∝ -K = -n, versus the density route's QK, unsuppressed.

With **β μ^2Λ_D^2/M^4 = 1** — the Lagrangian vanishes exactly at the DBI wall — the background law is closed-form (n ∝ a^{-3} exactly; ν n/(μ^2Λ_D)):

$$\frac{a_0^2(z)}{a_0^2(0)} = \frac{\sqrt{1+\nu_0^2}}{\sqrt{1+\nu_0^2(1+z)^6}}$$

constant to <1% through every MOND test (z ≤ 5), transition at z_t = ν_0^(-1/3) - 1 ∈ [17, 35], off at recombination (0.002–0.006). **w = -1 stays exact throughout** — the vacuum never rolls; the *trace* excitation (Ω ≤ 4.4 × 10^{-7}, the stage-3 black-hole ceiling) climbs the wall and its pressure reaches M^4 regardless of its amount, because the wall height is a property of the potential, not the charge. The window ν_0 ∈ [2.1 × 10^{-5}, 1.8 × 10^{-4}] is cut from *below* by the CMB off-switch and from *above* by the RAR — and the RAR ceiling exists only because the captured charge **drains** (stages 2–3): had it formed a halo, the local a$_0$ would fall to 0.55 of nominal at the g = a$_0$ radius, RAR-fatal. **The adverse drain result of non-claim 2d is load-bearing for this derivation.** The old CPL-dressed law (bump at z ≈ 0.5) is withdrawn; the derived law has no bump. Scripts: `nbody_2026/stage17_a0z_from_the_action_2026.py` (17/17), `nbody_2026/stage18_perturbation_matrix_with_mixing_2026.py` (10/10).

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2. What is verified

#	property	value / statement	script
1	**Lensing** — the test that killed modified inertia	$\Phi = \Psi \implies \gamma$_PPN = 1, M_dyn/M_lens = 1 exactly. **21.2**σ → **0.601**σ	`mi_relativistic_completion_aest_`
2	**Kernel embeds** in _Y	deep-MOND μ →x exact; Newtonian \$ →1; $x(u) a bijection (h = e^(-u)(1+u) > 0); free function **convex**|same||3| **Newtonian residual**|e^(-\sqrt{}$y) = 3.6×10^{-3457} at Earth's orbit	same
4	**Dark energy is not an input**	w = -1 **exactly** at the condensate minimum — proved, not assumed	`mi_condensate_vacuum_energy_a0_`
5	**Ghost-free** over the whole field range	K = μ^2(1-s^2)^(-3/2) > 0 for all |s|<1	`mi_dbi_khronon_2026.py`
6	**Subluminal** everywhere	c_s^2 = Λ_D s(1-s^2)/(1+Λ_D s) ≤ 0.385 Λ_D	same
7	**The published no-go is dissolved**	Blanchet & Skordis 2024 §4.3.1's 455× conflict **reverses sign**: bounded pressure ⟹ w\$→\$0 early ⟹ cosmology becomes a *lower* bound on μ^{-1}, same direction as MOND	same
8	**Theorem**: why no polynomial works	for K ~ u^n, early-time w → 1/(n-1). Quadratic 1, quartic $\frac{1}{3}$, sextic — all fail. **Only boundedness gives w\$→\$0**	same

#	property	value / statement	script
9	**CMB acoustic peaks** — real CLASS run	at Λ_D $\leq 10^{-2}$, c_s^2(recomb) = 2.9×10^{-8}; TT unchanged to **0.069%**, P(k=0.2) to 1.71%. Indistinguishable from CDM	`mi_dbi_cmb_class_run_2026.py`
10	**BTFR**	exact, from convexity	`mi_route_a_field_theory_2026.py`
11	**g^{-2} Lorentz violation**	**restored** by the aether (pure Bekenstein–Milgrom had lost it)	`mi_relativistic_completion_aest_`
12	**a_0 tied to Λ**	a_0 = m_cond/(4$\sqrt{}$$\pi$) with m_cond = M^2/(\$$\sqrt{}2M_Pl$), M^{4}\$ = ρ_Λ — agrees to **0.076%**	`mi_condensate_vacuum_energy_a0_2`
13	**Clusters** (v2)	~~shift-charge IC route~~ **withdrawn**: killed by the 1-Mpc confrontation — cluster R500 and galaxy-outskirt lensing share a scale with disjoint requirements, correlated Gaussian ICs cannot select environments, and **smooth accretion restores** **\$** $\to$1*foranycoldIC* ∗ ∗.*Replacedbythe* ∗ ∗*a**{0}**−bumpresponse*∗ ∗(*row*15)\|`*mi*$_i$*c*$_r$*oute*$_1$*mpc*$_c$*unfrozen*$_2$026.*py*`(9/9)||14|∗ ∗*DirectionalEFEsignal*∗ ∗|*pureMIpredicted* ∗ *exactlyzero*∗; *AQUAL*− *classpredicts*1–4|15| ∗ ∗*a***{0}\$-bump environment response**	passes all five environments: cluster (calibration, μ^2_eff = 0.23 Mpc^{-2}), galaxy interior 0.034% of M_b (RAR cost 4×10^{-4} dex), galaxy 1 Mpc 1.2% vs the 5.9% strict bound, linear cosmos 0.6% of mean matter, solar 10^{-19}. Vanishes on FRW (Y=0) and is second-order in perturbations ⇒ linear CMB/P(k) untouched by construction. Evades both prior kills (a response has no charge to advect; couples to environment directly)

#	property	value / statement	script
16	**...and its health check**	**no ghosts ever** (kinetic contribution 2AB ≥ 0, $= 0$ on FRW), no Ostrogradsky sector; FRW gradient health **excludes the old Λ_D $= 10^{-2}$ and forces Λ_D $\leq 8.4 \times 10^{-7}$** (window stays 3.7 orders); halo gradient health **caps A $\leq$ 2.7$\times$ fiducial, excluding the demanding end of Mistele's cluster band** — a two-sided falsifiable pinch; free signature: cluster anisotropic stress at O(0.6)	`mi_a0_bump_health_2026.py` (11/11)

#	property	value / statement	script
17	**...and the FULL perturbation matrix (v3)**	with all aether mixings: **c_T = 1 exact** (the new terms are algebraic in the metric — GW170817 safe by construction); **no-ghost is a THEOREM** (the bump's kinetic addition is rank-1 PSD in the $\chi\chi$ entry $\Longrightarrow$ Weyl monotonicity: it cannot lower any eigenvalue of AeST's healthy kinetic matrix; closed-form 2×2 + 200 random 4×4); unit-norm gives $\delta A^0 = -\Phi A^0 \Longrightarrow \delta Q = \chi - Q_0\Phi$ — the quasi-static phenomenology of row 15 is **derived, not assumed**; gradient block exact: ΔG = 2λ[[q, 2y^{3/2}B],[2y^{3/2}B, yq]], q = $(3y^2-8y+1)/(1+y)^4$; integrating out the aether softens the amplitude cap: at base_a = 1 the band is **A_max** $\in$ **[2.72, 4.46]**$\times$ **fiducial = [4.49, 7.36] Mpc**$^{-2}$ (**v4 correction:** the earlier "(2.7–7.4)$\times$ fiducial" spliced fiducial units with Mpc^{-2}, making the band look 1.66$\times$ wider at the top than it is — `base_a_lookup_and_robustness_2026.py`, 12/12). **The base_a lookup is now DONE and it converts the caveat rather than closing it:** base_a is built from AeST's single	`mi_aest_full_matrix_bump_2026.py` (11/11)

#	property	value / statement	script
18	**a_0(z) DERIVED from the action (v7)**	$a_0^2(Q) = \kappa^2 c^2 G(-K(Q))/c^2$ — the MOND scale is the dark sector's **pressure**. Zero new parameters; one relation $\beta = \mu^2 \Lambda\_D^2/M^4 = 1$ (**selected** by the CMB off-switch, not derived — non-claim 2e). Today's coefficient reproduced to the digit; **w = -1 stays exact** (the vacuum never rolls; the total pressure evolves). Density promotion **excluded by one bit** (a_0 would rise into the past); the exclusion theorem for the naive route is due to an adversarial external pass (Gemini, 2026-08-10) and is credited as such. Closed-form law: $a_0^2(z)/a_0^2(0) = \sqrt{(1+\nu_0^2)}/\sqrt{(1+\nu_0^2(1+z)^6)}$; constant to <1% for z $\leq$ 5; transition z_t $\in$ [17, 35]; off at recombination (0.002–0.006). Window $\nu_0 \in [2.1 \times 10^{-5}, 1.8 \times 10^{-4}]$, cut by the CMB from below and the RAR-via-drain from above — **non-claim 2d's adverse result is load-bearing here**. The CPL-dressed law (never self-consistent with w = -1 exact) is withdrawn	`stage17_a0z_from_the_action_2026` (17/17)

#	property	value / statement	script
19	**...and ITS full perturbation matrix (v7)**	the promotion's _YQ $\neq 0$ mixing analysed: **the cosmological matrix is untouched EXACTLY** ($h^{00} = g^{00} + A^0 A^0 = 0$ to all orders once unit-norm fixes $A^0 \Longrightarrow \delta Y^1 = 0$ identically; every new entry carries a positive power of y, and y = 0 on FRW — the CLASS pass, k^4-Jeans structure and GDM degeneracy theorem inherit without recomputation); **the no-ghost theorem TRANSFERS kernel-independently** (both kinetic additions land in the same $\chi\chi$ slot as row 17's: (F-yF) ≥ 0 because ghost-freedom of the base meets concavity of the kernel, and $(^2/)y^2F$ is a perfect square with $\propto$ -K = $M^4/\sqrt{(1+\nu^2)}$ > 0 always — a_0^2 can never cross zero; Weyl monotonicity does the rest); the mixing never reaches the kinetic matrix (δY has no time derivative), its drift dispersion is real at every k, and its gradient/force corrections are trace-bounded at $1.3 \times 10^{-4}/1.7 \times 10^{-4}$ of committed entries; c_T = 1 survives (still algebraic in the metric); vector WATCH inherited unchanged. Residual: WKB/committed-	`stage18_perturbation_matrix_with` (10/10)

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3. Parameter count, honestly

	ΛCDM	this completion
dark sector	*Ωc, ΩΛ*	M (= ρ_Λ^{1/4}), I_0, μ, Λ_D **(v7: Λ_D = M^2/μ, no longer free), A (v2)**
MOND scale	—	a_0 — **not independent**, = m_cond/(4√π) via item 12; **(v7) and its z-dependence is now derived too (§1.3)**
bounds	—	Λ_D: $1.9 \times 10^{-10} \ll$ Λ_D $\leq 8.4 \times 10^{-7}$ (**health-bounded, v2**; 3.7 orders) — **(v7) now fixed at M^2/μ by $\beta = 1$, which must land inside this window: a consistency check inherited rather than retired**; A ≈ 1.7 Mpc^{-2} calibrated on clusters, **health cap A ≤ 4.5 Mpc^{-2} = 2.7× fiducial** (row 16; the full matrix widens it to [2.72, 4.46]× fiducial = [4.49, 7.36] Mpc^{-2} at base_a = 1 — row 17, where the base_a fork and its robustness are stated); μ^{-1} = 4392 Mpc (item 12); I_0 ≈ Ω_dm (an IC); **(v7) the khronon's own trace charge $\nu_0 \in [2.1 \times 10^{-5}$, $1.8 \times 10^{-4}]$ (§1.3)**

So: **(v7) four dark-sector numbers against ΛCDM's two** (v2 added A; v7 removes Λ_D via $\beta = 1$), with a_0 — value *and* redshift dependence — derived from them rather than added. Still not fewer parameters than ΛCDM. What it buys is that Λ, dark matter, MOND, *and now the evolution of the MOND scale* come from **one function** instead of unrelated sectors.

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4. What is NOT claimed — read this before quoting anything above

1. $\kappa = \frac{1}{2}$ **IS NOT DERIVED.** Item 12 looks like a derivation and is not. `a  = m_cond/(4√ )` is *algebraically identical* to `a  = ½√(G _Λ)` and to `Z = 2√(8 /3)` — three costumes, one statement — and the residue is **convention-dependent** (4√π with the reduced Planck mass,

$\sqrt{}$2 with the non-reduced). A derived number cannot depend on a bookkeeping choice. It is a **relabelling**, as this corpus's own theorem already established. What genuinely moved: "why $\kappa = \frac{1}{2}$?" becomes "why is a_0 = m_cond/(4 $\sqrt{\pi}$)?", which is sharper because m_cond is a *derived* quantity of a real field theory rather than a definition.

2. **Dark matter EXISTS**, at the full Ω_dm. Removing the pressureless component moves H_3/H_1 by 54% and no refit absorbs it ($\Delta\chi^2 > 400$). What does **not** exist is a dark-matter *particle*: it is the Q-sector of the same scalar that supplies Λ and MOND. **(v3 correction:** v2 continued "and by item 13 it is absent where rotation curves are measured" — a dangling pointer, since v2 itself withdrew row 13. See 2d. **The defensible slogan shrinks to "no dark-matter particle" — never "no dark matter"; and (v5)** "none in galaxies" is WITHDRAWN as a claim of this theory**, not merely open — see 2d.)** 2d. **(v5) THE Q-SECTOR CANNOT ALSO BE ABSENT FROM GALAXIES — FALSIFIED WITHIN THE THEORY AS WRITTEN, AND THE REPAIR SPACE IS NOW PROVABLY EMPTY.** This is the largest change since v1 and it runs against the programme's interest. v2 asserted "none in galaxies" via a pointer to withdrawn row 13; a six-stage sequence (`nbody_2026/`, all scripts committed, 20/20 + 15/15 + 12/12 + 15/15
   - 15/15 + 11/11) then settled the question the pointer had concealed.

The physics. The 1-Mpc smooth-accretion theorem implies the Q-sector dust falls into *every* collapsing basin, galaxies included. What happens next was the open problem, and it closed in four steps:

- **Nothing stops the collapse.** Wave (k^4 ghost-condensate) pressure gives a soliton scale of **0.18 AU** at halo density — eight orders below the RAR region — and it *shrinks* as ρ grows ($\lambda \propto \rho$^-$\frac{1}{4}$); a 1 kpc wave core would need M below the Lyman-α fuzzy-DM floor. The dust *is* an n = 1 polytrope (p = $K\rho^2$, c_s^2 = $2K\rho$ rising) with a genuine **mass-independent 105 pc** equilibrium — but that lies **14.4× past DBI saturation**, where c_s → 0 and the stiffening switches off. Endpoint: a black hole of the captured share, **falsified** $5.8 \times 10^5\times$ **against Sgr A*** (4.3×10^6 M from individual stellar orbits — no mass model, kernel or cosmology in the comparison). *(v5 corrects v3/v4 here: an earlier draft claimed the DBI cap bounds the density and yields a ~250 pc core. $\mu^2 u^2/2$ is the **pressure**; exactly, ρ_exc = $Q_0\mu^2 u$ is **linear** in u and diverges at saturation. Withdrawn.)*
- **THEOREM: the dust mass IS the conserved shift charge.** $\rho = Q_0 \cdot n$ with n the shift-charge density, and **$\rho/n = Q_0$ independently of any Y-dependent Q-mass, of any shape or amplitude.** So no such term can change how much dust a galaxy carries — *a conserved charge cannot be suppressed locally, only moved (gravity moves it inward) or not conserved.* This single statement retro-explains three withdrawn mechanisms: the shift-charge IC route, the Φ^n response, and an acceleration-gated suppressor built and killed in stage 4–5 (its eight gates all tested the field amplitude u; the observable is n, and at fixed charge the suppressor makes a region a *cheaper* place to store charge — it attracts what it was built to expel).
- **Breaking the shift symmetry gets furthest, and fails.** A *global* potential is incompatible as an identity (dV/dt = Vφ must vanish for w = -1, while the condensate needs $\varphi = Q_0 \neq 0$); quantitatively the leak rates lock at $\Gamma n/\Gamma V = \rho\Lambda/\rho$dust = 2.60, so clearing a galaxy in 1 Gyr drifts $\rho\Lambda$ *by 47× and protecting* $\rho\Lambda$ forces a 180 Gyr charge lifetime. A **Y-gated** breaking, -W(Y/a_0^2)V(φ) with W(0) = 0, *defeats* that — on FRW the symmetry is **exact**, so cosmology is untouched identically. But **unlocking the charge does not unlock the energy**: ∇_μT^$\mu\nu$ = 0 holds regardless. **(v6 erratum:**

v5 added that the energy therefore *cannot leave*, bounding its export by the khronon's polytropic sound speed at ≈ 690 Gyr. That was a category error — a conservation law bounds total energy, not flux velocity; the causal limit is *c*, and row 17's tensor modes travel at exactly *c*. **Withdrawn.** What kills the export route instead is the *measured global budget*: a halo-gated conversion needs $\zeta \approx 0.6$ against the epoch-marginalised $\zeta < 0.0204$–0.0374 [McCarthy & Hill, PRD **108**, 063501] — over by 16–29× — leaving Ω_dr,0 ≈ 0.196, some 2100× today's radiation density; and dust→ Λ needs far more than $|\delta_0| \leq 6.7 \times 10^{-4}$. The gate is adverse rather than protective, since dilution goes as a_c.**)

- ** AND LENSING CLOSES IT BY A THEOREM.** Transforming the equation of state almost works: the dynamical source ρ + 3p vanishes at exactly **f = 1/3**, and the field-strength gate makes that a *stable fixed point* rather than a tuning. But **dynamics responds to ρ + 3p while lensing responds to ρ + p** — at f = 1/3 the lensing source is still $(2/3)\rho$, giving M_lens/M_dyn ≈ 29 against an observed 1.0–1.3. Generally, ρ + 3p = 0 requires w = -1/3 and ρ + p = 0 requires w = -1: **incompatible. No equation of state renders a given energy density invisible to both orbits and light; the only such configuration is $\rho = 0$**, which the transport bound forbids reaching. *The no-slip identity $\Phi = \Psi$ that makes this theory's modified-gravity arm work — the property that killed its modified-inertia arm at 21σ and underwrites item 1 — is the same property that forbids hiding the dust.*

Status, stated without spin. The galaxy-interior problem is not a missing mechanism: it is a conservation law (energy, via general covariance), plus a transport bound, plus lensing. **The repair space of local, energy-preserving modifications is provably empty.** The defensible slogan is therefore **"no dark-matter particle"** and nothing more; **"none in galaxies" is withdrawn as a claim of this theory.** What remains open is a structural change — the Q-sector not acquiring the galactic energy in the first place — against which the smooth-accretion theorem stands for any cold initial condition. Items 1–12 and 15–17 are untouched by all of this.

2e. **(v7) THE $a_0(z)$ DERIVATION'S OWN CAVEATS, stated before anyone quotes row 18 as finished.** (i) **β = 1 is SELECTED, not derived.** The relation μ^2Λ_D^2 = M^4 ("the Lagrangian vanishes at the DBI wall") is chosen because the CMB off-switch requires the floor (1-β) $\leq 3.6 \times 10^{-5}$; nothing in this paper derives it. A brane-tension/boundary argument would make it structural; until then it is one exact tuning carried openly — though it *removes* a parameter rather than adding one. (ii) **The DESI tension is not resolved and cannot be resolved by this sector**: the non-dust remainder's w(z) ≥ -1 always, so DESI's preferred phantom past (w_0 + w_a = -1.61) is unreachable. If DESI's evolving-w detection stands, it remains evidence against w = -1 exact. What v7 fixes is only that the $a_0(z)$ law no longer *depends* on borrowing DESI's numbers. (iii) **Owed**: the CLASS re-run with the derived law (row 19 proves the perturbation equations are identical and only the background $a_0(z)$ changes — expected to pass, but expected is not verified); the MUSE/MSA-3D re-examination (both were graded against the withdrawn CPL shape, whose z ≈ 0.5 bump is gone); a referee-grade covariant SVT decomposition behind row 19's WKB analysis.

3. **The post-recombination growth history is NOT verified.** Item 9 covers the acoustic peaks only. Holding c_s^2 at its peak value for all time *destroys* P(k=0.2), so the pass depends essentially on the bump being transient (it peaks at z ≈ 189). **A patched CLASS fluid carrying c_s^2(a) is required, not optional.**
4. **(v4 rewording — the original referred to withdrawn row 13.)** The dust component's small-scale initial conditions are **not confronted with Lyman-α.** The withdrawn IC route

needed a khronon transfer function $T \approx 0.33$ at $k \approx 4.5\ \mathrm{Mpc}^{-1}$ (ΛCDM-excluded); the surviving picture (standard cold dust + the bump response) should be Ly-α-safe by construction, but that is **asserted, not computed — the confrontation remains owed item in the programme.**

5. **Cassini Q_2 is inherited** (3–15σ) and is not relieved by anything here.
6. **PPN preferred-frame parameters** α_1, α_2 for this have not been computed against lunar-laser and binary-pulsar bounds.
7. **The wide-binary target is the point-field asymptote only.** γ_v = $\sqrt{\nu}$(y_extN); the full nonlinear AQUAL-EFE solve is owed.
8. **This is not a theory of everything** and no part of it addresses the Standard Model. The 2026-06-23 retraction stands.

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5. The falsifiable predictions

prediction	value	how to test
wide-binary boost	γ_v = $\sqrt{\nu}$(y_extN)	**Gaia DR4**, pre-registered and hash-stamped
directional EFE asymmetry	1–4%, signed	already firing at p = 0.029 with the right sign
cluster-to-cluster **scatter** in the residual	$\sigma(\xi)/\xi \approx$ 0.02–0.60	existing cluster samples — **(v4) separates the a_0-bump response (scatter tracks each cluster's distance from the resonance, rows 15–17) from any fixed-scale mechanism, which predicts uniformity** (the two prior candidates are withdrawn)
cluster mass **profile** misfit	ρ_c flat, M_c ~ r^3, can go negative	fit clusters with the khronon profile instead of NFW
g^{-2} Lorentz violation	computable s_$\mu\nu$	SME sector bounds
accelerated structure formation	earlier massive objects	JWST high-z massive galaxies (McGaugh et al. 2024)
a_0(z) shape (v7)	**constant to <1% for z ≤ 5, transition z_t ∈ [17, 35], NO bump at z ≈ 0.5**	any measurement of a_0 evolving below z ≈ 5 falsifies the derived law (the withdrawn CPL dressing predicted +6% at z = 0.5 — the two are now distinguishable); MUSE-type high-z kinematics, high-z RAR

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6. Credit

$\nu = \sqrt{}(1+1/y)$ is **Milgrom 1999 PLA 253:273 eq 9**. MOND: Milgrom 1983 ApJ 270:365. AQUAL: Bekenstein & Milgrom 1984 ApJ 286:7. TeVeS: Bekenstein 2004 PRD 70:083509. **AeST: Skordis & Złośnik 2021 PRL 127:161302** — the completion itself. Ghost condensate: Arkani-Hamed, Cheng, Luty & Mukohyama 2004 JHEP 0405:074; nonlinear: +Wiseman 2007 JHEP 0701:036. K(Q) = μ^2(Q-1)^2 and the ghost- condensate identification: Verwayen, Skordis & Złośnik 2024; **Blanchet & Skordis 2024 JCAP 11(2024)040** — including §4.3.1, the no-go this work dissolves. Quasi-static AeST: Durakovic & Skordis 2024 JCAP 04:040; Verwayen, Skordis & Boehm 2024 MNRAS 531:272. **Mistele, McGaugh & Hossenfelder 2023 A&A 676:A100** — the weak-lensing bounds that killed this program's first two cluster mechanisms and now two-sidedly bound the third (rows 15–17). GDM bound: Kopp, Skordis, Thomas & Ilić 2018 PRL 120:221102. Ly-α floor: Rogers & Peiris 2021 PRL 126:071302. RAR scatter: Desmond 2023. Accelerated structure formation: McGaugh, Schombert, Lelli & Franck 2024.

The distinctive content of this work is the coefficient — $a_0 = \kappa c \sqrt{}(G\rho\_\Lambda)$ with $\kappa = \frac{1}{2}$ — its embedding in _Y, the offset-DBI Q-sector, and items 7, 8, 9, 12 and 13. Everything else is cited above and is not mine.